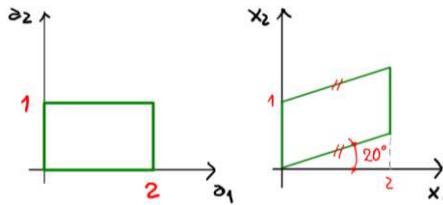


Nombre y Apellido:

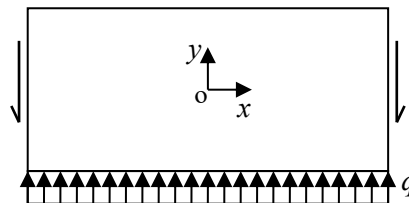
Firma:

Recuperatorio Segundo Examen Parcial 28/06/2024

- Dado el cuerpo sometido a deformaciones homogéneas como indicado en la figura, determinar:
 - El campo de desplazamientos.
 - El tensor de deformaciones de Green-Lagrange.
 - El tensor de deformaciones de Almansi.



- Una placa delgada $-L \leq x \leq L$, $-c \leq y \leq c$, de espesor unitario, compuesta de un material isotrópico elástico de tipo Hookeano ($\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2 \mu \varepsilon_{ij}$), con parámetros materiales homogéneos, está cargada como se indica en la figura. No existen fuerzas másicas y se asume hipótesis de pequeñas deformaciones.



Se propone la siguiente solución de tensiones:

$$\begin{aligned}\sigma_{xx} &= \frac{3}{4} \frac{q}{c^3} \left[\left(L^2 - \frac{2}{5} c^2 - x^2 \right) y + \frac{2}{3} y^3 \right] & \sigma_{yy} &= -\frac{3}{4} \frac{q}{c^3} \left(\frac{1}{3} y^3 - c^2 y + \frac{2}{3} c^3 \right) \\ \sigma_{xy} &= -\frac{3}{4} \frac{q}{c^3} (c^2 - y^2) x & \sigma_{xz} &= \sigma_{yz} = \sigma_{zz} = 0\end{aligned}$$

Se pide:

- Verificar equilibrio.
- Determinar componentes de deformación (considerar el problema en tres dimensiones).
- Verificar compatibilidad de deformaciones para el estado plano:

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} = 2 \frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y}$$

- Determinar si el siguiente campo velocidad es irrotacional y/o incompresible, justificar:

$$v_x = k \frac{\partial \theta}{\partial y} \quad v_y = -k \frac{\partial \theta}{\partial x} \quad v_z = 0$$

Donde $\theta = \theta(x, y)$ y k es una constante (independiente de las coordenadas del espacio). Además, ¿cuánto vale la tasa de deformaciones? y ¿qué nombre recibe el campo $k\theta$?

- Partiendo de la ecuación de balance de cantidad de movimiento angular demostrar que el tensor de tensiones debe ser simétrico, considerando la ecuación de movimiento y de conservación de masa.
- Sea el movimiento de un cierto medio continuo descrito por las ecuaciones:

$$x_1 = a_1 e^{-t}, \quad x_2 = a_2 e^t, \quad x_3 = a_3 + a_2 (e^{-t} - 1)$$

El campo vectorial A_i en dicho medio está dado por:

$$A_1(x_1, x_2, x_3, t) = e^{-t} (x_1 - 2x_2 + 3x_3)$$

$$A_2(x_1, x_2, x_3, t) = e^{-2t} (x_1 - 2x_2 + 3x_3)$$

$$A_3(x_1, x_2, x_3, t) = e^{-3t} (x_1 - 2x_2 + 3x_3)$$

Calcule la derivada material de A_i en ese medio (con el resultado final dado en descripción euleriana).

Recuperatorio Pencial 2

Ejercicio 1

Descripción de movimiento:

$$\underline{x} = \underline{x}(\underline{z}) \quad \begin{cases} x_1 = z_1 \\ x_2 = z_2 + \tan 20^\circ x_1 = z_2 + \tan 20^\circ z_1 \end{cases}$$

Descripción de movimiento inversa:

$$\underline{z} = \underline{z}(\underline{x}) \quad \begin{cases} z_1 = x_1 \\ z_2 = x_2 - \tan 20^\circ x_1 \end{cases}$$

a) Campo de desplazamiento: $u_i = x_i - z_i$

$$u = u(\underline{z}) \quad \begin{cases} u_1 = x_1 - z_1 = 0 \\ u_2 = x_2 - z_2 = \tan 20^\circ z_1 \end{cases}$$

Descripción Lagrangiana

$$u = u(\underline{x}) \quad \begin{cases} u_1 = 0 \\ u_2 = \tan 20^\circ x_1 \end{cases}$$

Descripción Euleriana

b) Tensor de deformación de Green-Lagrange

Alternativa 1:

$$E_{ij} = \frac{1}{2} \left(\frac{\partial x_k}{\partial a_i} \cdot \frac{\partial x_k}{\partial a_j} - \delta_{ij} \right)$$

$$\begin{aligned} E_{11} &= \frac{1}{2} \left(\frac{\partial x_1}{\partial a_1} \cdot \frac{\partial x_1}{\partial a_1} + \frac{\partial x_2}{\partial a_1} \cdot \frac{\partial x_2}{\partial a_1} - 1 \right) = \\ &= \frac{1}{2} \left(1 \cdot 1 + \tan 20^\circ \cdot \tan 20^\circ - 1 \right) = \frac{1}{2} \tan^2 20^\circ \approx 0,06624 \end{aligned}$$

$$E_{22} = \frac{1}{2} \left(\frac{\partial x_1}{\partial a_2} \cdot \frac{\partial x_1}{\partial a_2} + \frac{\partial x_2}{\partial a_2} \cdot \frac{\partial x_2}{\partial a_2} - 1 \right) =$$

$$= \frac{1}{2} (0 \cdot 0 + 1 \cdot 1 - 1) = 0 //$$

$$E_{12} = \frac{1}{2} \left(\frac{\partial x_1}{\partial a_1} \cdot \frac{\partial x_1}{\partial a_2} + \frac{\partial x_2}{\partial a_1} \cdot \frac{\partial x_2}{\partial a_2} \right) =$$

$$= \frac{1}{2} (1 \cdot 0 + \tan 20^\circ \cdot 1) = \frac{1}{2} \tan 20^\circ \approx 0,1820 //$$

Alternativa 2: $E_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial a_i} + \frac{\partial u_i}{\partial a_j} + \frac{\partial u_k}{\partial a_i} \cdot \frac{\partial u_k}{\partial a_j} \right)$

$$E_{11} = \frac{\partial u_1}{\partial a_1} + \frac{1}{2} \left(\frac{\partial u_1}{\partial a_1} \cdot \frac{\partial u_1}{\partial a_1} + \frac{\partial u_2}{\partial a_1} \cdot \frac{\partial u_2}{\partial a_1} \right) =$$

$$= 0 + \frac{1}{2} (0 \cdot 0 + \tan 20^\circ \cdot \tan 20^\circ) = \frac{1}{2} \tan^2 20^\circ //$$

$$E_{22} = \frac{\partial u_2}{\partial a_2} + \frac{1}{2} \left(\frac{\partial u_1}{\partial a_2} \cdot \frac{\partial u_1}{\partial a_2} + \frac{\partial u_2}{\partial a_2} \cdot \frac{\partial u_2}{\partial a_2} \right) =$$

$$= 0 + \frac{1}{2} (0 \cdot 0 + 0 \cdot 0) = 0 //$$

$$E_{12} = \frac{1}{2} \left(\frac{\partial u_2}{\partial a_1} + \frac{\partial u_1}{\partial a_2} + \frac{\partial u_1}{\partial a_1} \cdot \frac{\partial u_1}{\partial a_2} + \frac{\partial u_2}{\partial a_1} \cdot \frac{\partial u_2}{\partial a_2} \right) =$$

$$= \frac{1}{2} (\tan 20^\circ + 0 + 1 \cdot 0 + \tan 20^\circ \cdot 0) = \frac{1}{2} \tan 20^\circ //$$

c) Tensor de deformación de Almansi

Alternativa 1:
$$e_{ij} = \frac{1}{2} \left(\delta_{ij} - \frac{\partial a_k}{\partial x_i} \frac{\partial a_k}{\partial x_j} \right)$$

$$e_{11} = \frac{1}{2} \left(1 - \frac{\partial a_1}{\partial x_1} \frac{\partial a_1}{\partial x_1} - \frac{\partial a_2}{\partial x_1} \frac{\partial a_2}{\partial x_1} \right) = \frac{1}{2} \left[1 - 1 \cdot 1 - (-\tan 20^\circ)(-\tan 20^\circ) \right] = -\frac{1}{2} \tan^2 20^\circ \approx -0,06624 //$$

$$e_{22} = \frac{1}{2} \left(1 - \frac{\partial a_1}{\partial x_2} \frac{\partial a_1}{\partial x_2} - \frac{\partial a_2}{\partial x_2} \frac{\partial a_2}{\partial x_2} \right) = \frac{1}{2} \left(1 - 0 \cdot 0 - 1 \cdot 1 \right) = 0 //$$

$$e_{12} = \frac{1}{2} \left(-\frac{\partial a_1}{\partial x_1} \frac{\partial a_1}{\partial x_2} - \frac{\partial a_2}{\partial x_1} \frac{\partial a_2}{\partial x_2} \right) = \frac{1}{2} \left[-1 \cdot 0 - (-\tan 20^\circ) \cdot 1 \right] = \frac{1}{2} \tan 20^\circ \approx 0,1820 //$$

Alternativa 2:
$$e_{ij} = \frac{1}{2} \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} - \frac{\partial u_k}{\partial x_i} \frac{\partial u_k}{\partial x_j} \right)$$

$$e_{11} = \frac{\partial u_1}{\partial x_1} - \frac{1}{2} \left(\frac{\partial u_1}{\partial x_1} \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_1} \frac{\partial u_2}{\partial x_1} \right) = 0 - \frac{1}{2} (0 \cdot 0 + \tan 20^\circ \cdot \tan 20^\circ) = -\frac{1}{2} \tan^2 20^\circ //$$

$$e_{12} = \frac{\partial u_2}{\partial x_2} - \frac{1}{2} \left(\frac{\partial u_1}{\partial x_2} \frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_2} \frac{\partial u_2}{\partial x_2} \right) = 0 - \frac{1}{2} (0 \cdot 0 + 0 \cdot 0) = 0 //$$

$$\begin{aligned} e_{12} &= \frac{1}{2} \left(\frac{\partial u_2}{\partial x_1} + \frac{\partial u_1}{\partial x_2} - \frac{\partial u_1}{\partial x_1} \frac{\partial u_1}{\partial x_2} - \frac{\partial u_2}{\partial x_1} \frac{\partial u_2}{\partial x_2} \right) = \\ &= \frac{1}{2} \left(\tan 20^\circ + 0 - 0 \cdot 0 - \tan 20^\circ \cdot 0 \right) = \frac{1}{2} \tan 20^\circ // \end{aligned}$$

Exercício 2

a)

$$\frac{\partial V_{ji}}{\partial x_i} + \cancel{b_j^0} = 0$$

$$\frac{\partial V_{xx}}{\partial x} + \frac{\partial V_{xy}}{\partial y} + \frac{\partial V_{xz}}{\partial z}$$

$$\frac{3}{4} \frac{q}{c^3} (-2x)y + -\frac{3}{4} \frac{q}{c^3} (-2y)x = 0 \Rightarrow \text{Verifica equilibrio en } x$$

$$\frac{\partial V_{yx}}{\partial x} + \frac{\partial V_{yy}}{\partial y} + \frac{\partial V_{yz}}{\partial z}$$

$$-\frac{3}{4} \frac{q}{c^3} (c^2 - y^2) + -\frac{3}{4} \frac{q}{c^3} \left(\frac{2}{3} y^2 - c^2 \right) = 0 \Rightarrow \text{Verifica equilibrio en } y$$

$$\frac{\partial V_{zx}}{\partial x} + \frac{\partial V_{zy}}{\partial y} + \frac{\partial V_{zz}}{\partial z} = 0 \Rightarrow \text{Verifica equilibrio en } z$$

b)

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij} \quad (1)$$

$$\sigma_{mm} = \lambda \epsilon_{kk} \delta_{mm} + 2\mu \epsilon_{mm} \Rightarrow \epsilon_{kk} = \frac{\sigma_{kk}}{3\lambda + 2\mu} \quad (2)$$

$$(1) \text{ y } (2) \Rightarrow \epsilon_{ij} = \frac{1}{2\mu} \left(\sigma_{ij} - \frac{\lambda}{3\lambda + 2\mu} \sigma_{kk} \delta_{ij} \right)$$

$$\begin{aligned} \sigma_{kk} &= \sigma_{xx} + \sigma_{yy} + \cancel{\sigma_{zz}^0} = \frac{3}{4} \frac{q}{c^3} \left[\left(L^2 - \frac{2}{5} c^2 + c^2 - x^2 \right) y + \frac{2}{3} y^3 - \frac{1}{3} y^3 - \frac{2}{3} c^3 \right] \\ &= \frac{3}{4} \frac{q}{c^3} \left[\left(L^2 + \frac{3}{5} c^2 - x^2 \right) y + \frac{1}{3} y^3 - \frac{2}{3} c^3 \right] \end{aligned}$$

$$\begin{aligned} \epsilon_{xx} &= \frac{1}{2\mu} \left(\cancel{\tau_{xx}} - \frac{\lambda}{3\lambda+2\mu} \tau_{kk} \overset{1}{\sigma_{xx}} \right) = \\ &= \frac{1}{2\mu} \frac{3}{4} \frac{q}{c^3} \left\{ \left(\frac{L^2 - \frac{2}{5} c^2 - x^2}{5} \right) y + \frac{2}{3} y^3 - \frac{\lambda}{3\lambda+2\mu} \left[\left(L^2 + \frac{3}{5} c^2 - x^2 \right) y + \frac{1}{3} y^3 - \frac{2}{3} c^3 \right] \right\} // \end{aligned}$$

$$\begin{aligned} \epsilon_{yy} &= \frac{1}{2\mu} \left(\tau_{yy} - \frac{\lambda}{3\lambda+2\mu} \tau_{kk} \right) = \\ &= \frac{1}{2\mu} \frac{3}{4} \frac{q}{c^3} \left\{ -\frac{1}{3} y^3 + c^2 y - \frac{2}{3} c^3 - \frac{\lambda}{3\lambda+2\mu} \left[\left(L^2 + \frac{3}{5} c^2 - x^2 \right) y + \frac{1}{3} y^3 - \frac{2}{3} c^3 \right] \right\} // \end{aligned}$$

$$\begin{aligned} \epsilon_{xy} &= \frac{1}{2\mu} \left(\tau_{xy} - \frac{\lambda}{3\lambda+2\mu} \tau_{kk} \overset{0}{\sigma_{xy}} \right) = \\ &= \frac{1}{2\mu} \left(-\frac{3}{4} \right) \frac{q}{c^3} (c^2 - y^2) x // \end{aligned}$$

$$\begin{aligned} \epsilon_{zz} &= \frac{1}{2\mu} \left(\cancel{\tau_{zz}} - \frac{\lambda}{3\lambda+2\mu} \tau_{kk} \right) = \\ &= \frac{1}{2\mu} \frac{(-\lambda)}{3\lambda+2\mu} \left[\left(L^2 + \frac{3}{5} c^2 - x^2 \right) y + \frac{1}{3} y^3 - \frac{2}{3} c^3 \right] // \end{aligned}$$

$$\epsilon_{xz} = \frac{1}{2\mu} \left(\cancel{\tau_{xz}} - \frac{\lambda}{3\lambda+2\mu} \tau_{kk} \cdot 0 \right) = 0 //$$

$$\epsilon_{yz} = \frac{1}{2\mu} \left(\cancel{\tau_{yz}} - \frac{\lambda}{3\lambda+2\mu} \tau_{kk} \cdot 0 \right) = 0 //$$

$$c) \quad \frac{\partial^2 E_{xx}}{\partial y^2} + \frac{\partial^2 E_{yy}}{\partial x^2} = 2 \frac{\partial^2 E_{xy}}{\partial x \partial y}$$

$$\frac{\partial^2 E_{xx}}{\partial y^2} = \frac{1}{2\mu} \frac{3}{4} \frac{q}{c^3} \left[4y - \frac{\lambda}{3\lambda + 2\mu} 2y \right] = \frac{3}{4} \frac{q}{c^3} \frac{y}{\mu} \frac{5\lambda + 4\mu}{3\lambda + 2\mu}$$

$$\frac{\partial^2 E_{yy}}{\partial x^2} = \frac{1}{2\mu} \frac{3}{4} \frac{q}{c^3} \frac{\lambda}{3\lambda + 2\mu} \cdot 2y$$

$$\frac{\partial^2 E_{xy}}{\partial x \partial y} = \frac{1}{2\mu} \left(-\frac{3}{4} \right) \frac{q}{c^3} (-2y) = \frac{3}{4} \frac{q}{c^3} \frac{y}{\mu}$$

$$\frac{3}{4} \frac{q}{c^3} \frac{y}{\mu} \frac{5\lambda + 4\mu}{3\lambda + 2\mu} + \frac{3}{4} \frac{q}{c^3} \frac{y}{\mu} \frac{\lambda}{3\lambda + 2\mu} = 2 \cdot \frac{3}{4} \frac{q}{c^3} \frac{y}{\mu}$$

$$5\lambda + 4\mu + \lambda = 2(3\lambda + 2\mu)$$

$$6\lambda + 4\mu = 6\lambda + 4\mu$$

$0 = 0 \rightarrow$ Verifica la ecuación de compatibilidad

Ejercicio 3:

$$\theta = \theta(x, y)$$

Irrotacional: $\nabla \times \underline{v} = \underline{0}$

$$\nabla \times \underline{v} = \begin{vmatrix} k \frac{\partial \theta(x, y)}{\partial y} & -k \frac{\partial \theta(x, y)}{\partial x} & 0 \\ \underline{e}_x & \underline{e}_y & \underline{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} = -k \frac{\partial^2 \theta}{\partial x^2} \underline{e}_z - k \frac{\partial \theta}{\partial y^2} \underline{e}_z =$$

$$= \left(-k \frac{\partial^2 \theta}{\partial x^2} - k \frac{\partial \theta}{\partial y^2} \right) \underline{e}_z$$

En general no es irrotacional, para solo se tiene que cumplir que:

$$-k \left(\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial \theta}{\partial y^2} \right) = 0 \quad \text{si } k \neq 0$$

Incompresible: $\text{div}(\underline{v}) = 0$

$$k \frac{\partial^2 \theta}{\partial x \partial y} + (-k) \frac{\partial^2 \theta}{\partial y \partial x} + 0 = 0 \Rightarrow \text{independiente de } \theta, \text{ el flujo es incompresible}$$

Tasa de deformación:

$$v_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right)$$

$$v_{xx} = \frac{\partial v_x}{\partial x} = k \frac{\partial^2 \theta}{\partial x \partial y}$$

$$v_{yy} = \frac{\partial v_y}{\partial y} = -k \frac{\partial^2 \theta}{\partial x \partial y}$$

$$v_{zz} = \frac{\partial v_z}{\partial z} = 0$$

$$v_{xy} = \frac{1}{2} \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) = \frac{1}{2} \left(k \frac{\partial \theta}{\partial y^2} - k \frac{\partial \theta}{\partial x^2} \right) = \frac{k}{2} \left(\frac{\partial \theta}{\partial y^2} - \frac{\partial \theta}{\partial x^2} \right)$$

$$v_{xz} = \frac{1}{2} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) = 0 //$$

$$v_{yz} = \frac{1}{2} \left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right) = 0 //$$

Los curvas de nivel de campo $K\theta(x,y)$ representan las líneas de corriente del flujo.

Ejercicio 4:

Cantidad de movimiento angular: $\mathcal{L}_i = \int_V \epsilon_{ijk} x_j \rho v_k dV$

Momento de los fuerzas externas: $\dot{\mathcal{L}}_i = \int_V \epsilon_{ijk} x_j b_k dV + \int_S \epsilon_{ijk} x_j T_k dS$

Balance de la cantidad de movimiento angular: $\frac{D}{Dt} \mathcal{L}_i = \dot{\mathcal{L}}_i$

Fórmula de Cauchy: $T_i = n_j \tau_{ji}$

Ecuación de movimiento: $\rho \frac{Dv_i}{Dt} = \frac{\partial \tau_{ij}}{\partial x_j} + b_i$

Ecuación de continuidad: $\frac{D\rho}{Dt} + \rho \frac{\partial v_k}{\partial x_k} = 0$

$$\begin{aligned} \int_S \epsilon_{ijk} x_j T_k dS &= \int_S \epsilon_{ijk} x_j n_k \tau_{ek} dS = \int_V \frac{\partial}{\partial x_k} (\epsilon_{ijk} x_j \tau_{ek}) dV = \\ &= \int_V \epsilon_{ijk} \left(\frac{\partial x_j}{\partial x_k} \tau_{ek} + x_j \frac{\partial \tau_{ek}}{\partial x_k} \right) dV \Rightarrow \end{aligned}$$

$$\Rightarrow \dot{\mathcal{L}}_i = \int_V \left[\epsilon_{ijk} x_j \left(b_k + \frac{\partial \tau_{ek}}{\partial x_k} \right) + \epsilon_{ijk} \tau_{jk} \right] dV$$

$$\begin{aligned} \frac{D}{Dt} \mathcal{L}_i &= \frac{D}{Dt} \int_V \epsilon_{ijk} x_j \rho v_k dV = \int_V \left[\frac{D}{Dt} (\epsilon_{ijk} x_j \rho v_k) + \epsilon_{ijk} x_j \rho v_k \frac{\partial v_k}{\partial x_k} \right] dV = \\ &= \int_V \left[\underbrace{\epsilon_{ijk} \frac{Dx_j}{Dt} \rho v_k}_{=0} + \epsilon_{ijk} x_j \frac{D\rho}{Dt} v_k + \epsilon_{ijk} x_j \rho \frac{Dv_k}{Dt} + \epsilon_{ijk} x_j \rho v_k \frac{\partial v_k}{\partial x_k} \right] dV = \end{aligned}$$

$$\epsilon_{ijk} \rho v_j v_k = \rho \cdot \frac{1}{2} (\epsilon_{ijk} v_j v_k - \epsilon_{ikj} v_k v_j) = 0$$

$$= \int_V \epsilon_{ijk} x_j v_k \underbrace{\left(\frac{D\rho}{Dt} + \rho \frac{\partial v_e}{\partial x_e} \right)}_0 dV + \int_V \epsilon_{ijk} x_j \rho \frac{Dv_k}{Dt} dV \Rightarrow$$

Por conservación de masa

$$\Rightarrow \frac{Dh_i}{Dt} = h_i \Rightarrow$$

$$\int_V \epsilon_{ijk} x_j \rho \frac{Dv_k}{Dt} dV = \int_V \left[\epsilon_{ijk} x_j \left(b_k + \frac{\partial \tau_{ek}}{\partial x_e} \right) + \epsilon_{ijk} \tau_{jk} \right] dV$$

$$\int_V \epsilon_{ijk} x_j \underbrace{\left(b_k + \frac{\partial \tau_{ek}}{\partial x_e} - \rho \frac{Dv_k}{Dt} \right)}_0 dV + \int_V \epsilon_{ijk} \tau_{jk} dV = 0 \Rightarrow$$

0 Por ec. de Movimiento

$$\Rightarrow \int_V \epsilon_{ijk} \tau_{jk} dV = 0 \quad \forall V \Rightarrow$$

$$\epsilon_{ijk} \tau_{jk} = 0$$

Si $i=j$, $i=k$ o $j=k \Rightarrow$ no opera a la simetría de la ecuación anterior

Entonces en el caso de $i \neq j$, $i \neq k$ y $j \neq k$, que son los términos no nulos que operan a la simetría $\Rightarrow$

$$\left. \begin{aligned} \overset{1}{e_{123}} \tau_{23} + \overset{-1}{e_{132}} \tau_{32} &= 0 \Rightarrow \tau_{23} = \tau_{32} \\ \overset{-1}{e_{213}} \tau_{13} + \overset{1}{e_{231}} \tau_{31} &= 0 \Rightarrow \tau_{13} = \tau_{31} \\ \overset{1}{e_{312}} \tau_{12} + \overset{-1}{e_{321}} \tau_{21} &= 0 \Rightarrow \tau_{12} = \tau_{21} \end{aligned} \right\} \Rightarrow \tau_{ij} \text{ es simétrico} \Rightarrow \tau_{ij} = \tau_{ji}$$

Exercício 5.

$$x_1 = a_1 \cdot e^{-t}$$

↓

$$a_1 = \frac{x_1}{e^{-t}} = x_1 e^t$$

$$x_2 = a_2 \cdot e^t$$

↓

$$a_2 = \frac{x_2}{e^t} = x_2 \cdot e^{-t}$$

$$x_3 = a_3 + a_2(e^{-t} - 1)$$

↓

$$a_3 = x_3 - a_2(e^{-t} - 1) =$$

$$a_3 = x_3 - x_2 \cdot e^{-t} \cdot (e^{-t} - 1)$$

Alternativa 1:

$$v_1 = \frac{D}{Dt}(x_1) = -a_1 e^{-t}$$

$$v_2 = \frac{D}{Dt}(x_2) = a_2 \cdot e^t$$

$$v_3 = \frac{D}{Dt}(x_3) = -a_2 \cdot e^{-t}$$

$$\Rightarrow \underline{v} = \begin{bmatrix} -a_1 \cdot e^{-t} \\ a_2 \cdot e^t \\ -a_2 \cdot e^{-t} \end{bmatrix} = \begin{bmatrix} -x_1 \\ x_2 \\ -x_2 e^{-2t} \end{bmatrix}$$

$a_i = a_i(x, t)$

$$A_i = e^{-\alpha_i t} (x_1 - 2x_2 + 3x_3)$$

$$\text{com } \alpha_1 = 1, \alpha_2 = 2 \wedge \alpha_3 = 3$$

$$A_i = e^{-\alpha_i t} \beta_j x_j$$

$$\text{ou } \beta_1 = \alpha_1, \beta_2 = -\alpha_2 \wedge \beta_3 = \alpha_3$$

$$\frac{\partial}{\partial t} A_i = -\alpha_{(i)} e^{-\alpha_i t} \beta_j x_j$$

→ sem convenção de soma

$$\frac{\partial}{\partial x_k} A_i = e^{-\alpha_i t} \beta_j \cdot \frac{\partial x_j}{\partial x_k} = \beta_k \cdot e^{-\alpha_i t}$$

$$\frac{DA_i}{Dt} = \frac{\partial A_i}{\partial t} + \frac{\partial A_i}{\partial x_j} \cdot v_j =$$

$$= -\alpha_{(i)} e^{-\alpha_i t} \beta_j x_j + \beta_j e^{-\alpha_i t} \cdot v_j$$

$$= e^{-\alpha_i t} \beta_j (v_j - \alpha_{(i)} x_j)$$

$$P_j(v_j - \alpha_{(i)} x_j) = [P_1, P_2, P_3] \cdot \begin{bmatrix} v_1 - \alpha_{(i)} x_1 \\ v_2 - \alpha_{(i)} x_2 \\ v_3 - \alpha_{(i)} x_3 \end{bmatrix} = [1, -2, 3] \cdot \begin{bmatrix} -x_1(1 + \alpha_{(i)}) \\ x_2(1 - \alpha_{(i)}) \\ -x_2 e^{-2t} - \alpha_{(i)} x_3 \end{bmatrix} =$$

$$= -x_1(1 + \alpha_{(i)}) + 2x_2(\alpha_{(i)} - 1) - 3x_2 e^{-2t} - 3\alpha_{(i)} x_3 =$$

$$= -x_1(1 + \alpha_{(i)}) + x_2(2\alpha_{(i)} - 2 - 3e^{-2t}) - 3\alpha_{(i)} x_3$$

$$P_j(v_j - \alpha_1 x_j) = -2x_1 - 3x_2 e^{-2t} - 3x_3$$

$$P_j(v_j - \alpha_2 x_j) = -3x_1 + x_2(2 - 3e^{-2t}) - 6x_3$$

$$P_j(v_j - \alpha_3 x_j) = -4x_1 + x_2(4 - 3e^{-2t}) - 9x_3$$

$$\frac{DA_1}{Dt} = e^{-t} \cdot (-2x_1 - 3x_2 e^{-2t} - 3x_3) //$$

$$\frac{DA_2}{Dt} = e^{-2t} \cdot [-3x_1 + x_2(2 - 3e^{-2t}) - 6x_3] //$$

$$\frac{DA_3}{Dt} = e^{-2t} \cdot [-4x_1 + x_2(4 - 3e^{-2t}) - 9x_3] //$$

Alternativa 2:

$$\underline{v} = \begin{bmatrix} -x_1 \\ x_2 \\ -x_2 e^{-2t} \end{bmatrix}$$

$$\frac{\partial A_1}{\partial t} = -e^{-t} (x_1 - 2x_2 + 3x_3)$$

$$\frac{\partial A_2}{\partial t} = -2e^{-2t} (x_1 - 2x_2 + 3x_3) \quad \Rightarrow \quad \frac{\partial A}{\partial t} = (x_1 - 2x_2 + 3x_3) \cdot \begin{bmatrix} -e^{-t} \\ -2e^{-2t} \\ -3e^{-3t} \end{bmatrix}$$

$$\frac{\partial A_3}{\partial t} = -3e^{-3t} (x_1 - 2x_2 + 3x_3)$$

$$A_i = e^{-\alpha_i t} (x_1 - 2x_2 + 3x_3) \quad \text{com } \alpha_1 = 1, \alpha_2 = 2 \wedge \alpha_3 = 3$$

$$\begin{aligned} \frac{\partial A_i}{\partial x_1} &= e^{-\alpha_i t} \\ \frac{\partial A_i}{\partial x_2} &= -2e^{-\alpha_i t} \\ \frac{\partial A_i}{\partial x_3} &= 3e^{-\alpha_i t} \end{aligned} \quad \Rightarrow \quad \underline{A} = \begin{bmatrix} \left[\frac{\partial A_i}{\partial x_1} \right] & \left[\frac{\partial A_i}{\partial x_2} \right] & \left[\frac{\partial A_i}{\partial x_3} \right] \end{bmatrix} = \begin{bmatrix} e^{-t} & -2e^{-t} & 3e^{-t} \\ e^{-2t} & -2e^{-2t} & 3e^{-2t} \\ e^{-3t} & -2e^{-3t} & 3e^{-3t} \end{bmatrix}$$

$$\frac{DA_i}{Dt} = \frac{\partial A_i}{\partial t} + \frac{\partial A_i}{\partial x_j} \cdot v_j$$

$$\frac{DA}{Dt} = (x_1 - 2x_2 + 3x_3) \cdot \begin{bmatrix} -e^{-t} \\ -2e^{-2t} \\ -3e^{-3t} \end{bmatrix} + \begin{bmatrix} e^{-t} & -2e^{-t} & 3e^{-t} \\ e^{-2t} & -2e^{-2t} & 3e^{-2t} \\ e^{-3t} & -2e^{-3t} & 3e^{-3t} \end{bmatrix} \begin{bmatrix} -x_1 \\ x_2 \\ -x_2 e^{-2t} \end{bmatrix}$$

$$= (x_1 - 2x_2 + 3x_3) \cdot \begin{bmatrix} -e^{-t} \\ -2e^{-2t} \\ -3e^{-3t} \end{bmatrix} + \begin{bmatrix} -x_1 e^{-t} - 2x_2 e^{-t} - 3x_2 e^{-t} \cdot e^{-2t} \\ -x_1 e^{-2t} - 2x_2 e^{-2t} - 3x_2 e^{-2t} \cdot e^{-2t} \\ -x_1 e^{-3t} - 2x_2 e^{-3t} - 3x_2 e^{-3t} \cdot e^{-2t} \end{bmatrix}$$

$$= \begin{bmatrix} (-2x_1 - 3x_2 e^{-2t} - 3x_3) e^{-t} \\ (-3x_1 + 2x_2 - 3x_2 e^{-2t} - 6x_3) e^{-2t} \\ (-4x_1 + 4x_2 - 3x_2 e^{-3t} - 9x_3) e^{-3t} \end{bmatrix}$$

Alternativa 3:

$$A_i = e^{-\alpha_i t} (x_1 - 2x_2 + 3x_3) \quad \text{con } \alpha_1 = 1, \alpha_2 = 2 \wedge \alpha_3 = 3$$

$\downarrow x_i = x_i(a, t)$

$$A_i = e^{-\alpha_i t} \left\{ a_1 \cdot e^{-t} - 2a_2 e^t + 3[a_3 + a_2(e^{-t} - 1)] \right\}$$

$$A_i = e^{-\alpha_i t} [a_1 \cdot e^{-t} + a_2(3e^{-t} - 3 - 2e^t) + 3a_3]$$

$\rightarrow$ sin convención de suma

$$\frac{DA_i(a, t)}{Dt} = -\alpha_i \cdot e^{-\alpha_i t} [a_1 \cdot e^{-t} + a_2(3e^{-t} - 3 - 2e^t) + 3a_3] +$$

$$+ e^{-\alpha_i t} \cdot [-a_1 \cdot e^{-t} + a_2(-3e^{-t} - 2e^t)] =$$

$$= -e^{-\alpha_i t} \left\{ a_1(1 + \alpha_i) \cdot e^{-t} + a_2[3e^{-t}(1 + \alpha_i) + 2e^t(1 - \alpha_i) - 3\alpha_i] + a_3 3\alpha_i \right\} =$$

Se reemplaza con $a_i = a_i(x, t)$:

$$\frac{DA_i(x, t)}{Dt} = -e^{-\alpha_i t} \left\{ x_1 \cdot e^{-t}(1 + \alpha_i) + x_2 e^{-t} [3e^{-t}(1 + \alpha_i) + 2e^t(1 - \alpha_i) - 3\alpha_i] + [x_3 - x_2 \cdot e^t(e^{-t} - 1)] 3\alpha_i \right\} =$$

$$= -e^{-\alpha_i t} \left\{ x_1(1 + \alpha_i) + x_2 e^{-t} [3e^{-t}(1 + \alpha_i) - 3(e^{-t} - 1)\alpha_i] + 2e^t(1 - \alpha_i) - 3\alpha_i \right\} + 3x_3 \alpha_i =$$

$$= -e^{-\alpha_i t} \left\{ x_1(1 + \alpha_i) + x_2 [3e^{-2t} + 2(1 - \alpha_i)] + 3x_3 \alpha_i \right\}$$

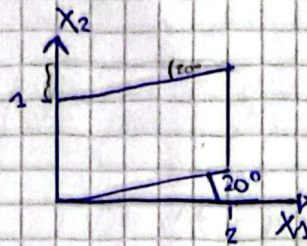
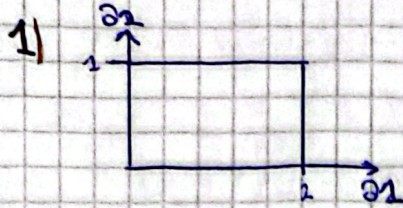
$$\frac{DA_1(x, t)}{Dt} = -e^{-t} (2x_1 + x_2 3e^{-2t} + 3x_3) //$$

$$\frac{DA_2(x, t)}{Dt} = -e^{-2t} [3x_1 + x_2(3e^{-2t} - 2) + 6x_3] //$$

$$\frac{DA_3(x, t)}{Dt} = -e^{-3t} [4x_1 + x_2(3e^{-2t} - 4) + 9x_3] //$$

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$$\operatorname{Tg} 20^\circ = 1$$



a) El campo de desplazamientos

Desplazamiento de nodos

$$\begin{cases} X_1 = a_1 \\ X_2 = a_2 + \operatorname{Tg} 20^\circ X_1 \end{cases} \rightarrow \begin{cases} a_1 \\ a_2 + \operatorname{Tg} 20^\circ a_1 \end{cases}$$

Campo de desplazamientos $U_i = X_i - a_i$

$$\begin{cases} U_1 = X_1 - a_1 \rightarrow a_1 - a_1 = 0 \\ U_2 = X_2 - a_2 \rightarrow (a_2 + \operatorname{Tg} 20^\circ a_1) - a_2 \rightarrow \operatorname{Tg} 20^\circ a_1 \end{cases}$$

b) El tensor de deformaciones de Green Lagrange

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial X_i}{\partial a_j} + \frac{\partial X_j}{\partial a_i} - S_{ij} \right)$$

$$S_{ij} = 1 \quad \forall i=j$$

$$S_{ij} = 0 \quad \forall i \neq j$$

$$\varepsilon_{11} = \frac{1}{2} \left(\frac{\partial X_1}{\partial a_1} + \frac{\partial X_1}{\partial a_1} - S_{11} \right) = \frac{1}{2} (1 + 1 - 1) = \frac{1}{2}$$

$$\varepsilon_{22} = \frac{1}{2} \left(\frac{\partial X_2}{\partial a_2} + \frac{\partial X_2}{\partial a_2} - S_{22} \right) = \frac{1}{2} (1 + 1 - 1) = \frac{1}{2}$$

$$\varepsilon_{12} = \varepsilon_{21} = \frac{1}{2} \left(\frac{\partial X_1}{\partial a_2} + \frac{\partial X_2}{\partial a_1} - S_{12} \right)$$

$$\frac{1}{2} (1 \cdot 0 + \operatorname{Tg} 20^\circ \cdot 1 - 0) = \frac{1}{2} \operatorname{Tg} 20^\circ$$

$$\varepsilon_{22} = \frac{1}{2} \left(\frac{\partial X_2}{\partial a_2} + \frac{\partial X_2}{\partial a_2} - S_{22} \right) =$$

$$\frac{1}{2} (0 \cdot 0 + 1 \cdot 1 - 1) = 0$$

$$\varepsilon_{ij} = \begin{bmatrix} \frac{1}{2} \operatorname{Tg}^2 20^\circ & \frac{1}{2} \operatorname{Tg} 20^\circ \\ \frac{1}{2} \operatorname{Tg} 20^\circ & 0 \end{bmatrix}$$

c) Tensor de deformaciones Almansi

$$\varepsilon_{ij} = \frac{1}{2} \left(S_{ij} - \frac{\partial a_k}{\partial x_i} \cdot \frac{\partial a_k}{\partial x_j} \right)$$

$$\varepsilon_{11} = \frac{1}{2} \left(S_{11} - \frac{\partial a_1}{\partial x_1} \frac{\partial a_1}{\partial x_1} - \frac{\partial a_2}{\partial x_1} \frac{\partial a_2}{\partial x_1} \right) = \frac{1}{2} \left(1 - 1 \cdot 1 - (-\operatorname{tg} 20^\circ)(-\operatorname{tg} 20^\circ) \right) = -\frac{1}{2} \operatorname{tg}^2 20^\circ$$

$$\varepsilon_{22} = \frac{1}{2} \left(S_{22} - \frac{\partial a_1}{\partial x_2} \frac{\partial a_1}{\partial x_2} - \frac{\partial a_2}{\partial x_2} \frac{\partial a_2}{\partial x_2} \right) = \frac{1}{2} \left(1 - 0 \cdot 0 - 1 \cdot 1 \right) = 0$$

$$\varepsilon_{12} = \frac{1}{2} \left(S_{12} - \frac{\partial a_1}{\partial x_1} \frac{\partial a_1}{\partial x_2} - \frac{\partial a_2}{\partial x_1} \frac{\partial a_2}{\partial x_2} \right) = \frac{1}{2} \operatorname{tg} 20^\circ$$

2 a) Verificar equilibrio

$$\sigma_{xx} = \frac{3}{4} \frac{q}{C^3} \left[\left(L^2 - \frac{2}{5} C^2 - x^2 \right) y + \frac{2}{3} y^3 \right]$$

$$\sigma_{yy} = -\frac{3}{4} \frac{q}{C^3} \left(\frac{1}{3} y^3 - C^2 y + \frac{2}{3} C^3 \right)$$

$$\sigma_{xy} = -\frac{3}{4} \frac{q}{C^3} (C^2 - y^2) x \quad \sigma_{xz} = \sigma_{yz} = \sigma_{zz} = 0$$

$$\textcircled{1} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} + b_1 = 0 \rightarrow \frac{3}{4} \frac{q}{C^3} (-2xy) - \left(\frac{3}{4} \frac{q}{C^3} 2yx \right) = 0 \checkmark$$

$$\textcircled{2} \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} + b_2 = 0 \rightarrow -\frac{3}{4} \frac{q}{C^3} (C^2 - y^2) - \frac{3}{4} \frac{q}{C^3} \left(\frac{1}{3} y^3 - C^2 y + \frac{2}{3} C^3 \right) = 0 \checkmark$$

$$\textcircled{3} \frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + b_3 = 0 \quad 0 + 0 + 0 + 0 = 0 \checkmark$$

Equilibrio

b) Determinar componentes de deformación: $\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}$ ^① integrar

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij} \quad \text{tomando } i, j = k$$

$$\hookrightarrow \epsilon_{kk} = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}$$

$$\sigma_{kk} = \lambda \epsilon_{kk} \delta_{kk} + 2\mu \epsilon_{kk}$$

$$\sigma_{kk} = \lambda \epsilon_{kk} 3 + 2\mu \epsilon_{kk} \rightarrow \left[\epsilon_{kk} = \frac{\sigma_{kk}}{3\lambda + 2\mu} \right] \text{ reemplazamos } \textcircled{1}$$

$$\sigma_{ij} = \lambda \left(\frac{\sigma_{kk}}{3\lambda + 2\mu} \right) \delta_{ij} + 2\mu \epsilon_{ij} \quad \text{Algebra}$$

$$\left[\epsilon_{ij} = \frac{1}{2\mu} \sigma_{ij} - \frac{\lambda}{2\mu(3\lambda + 2\mu)} \sigma_{kk} \delta_{ij} \right] \text{ como planes calculamos los 6 componentes}$$

$$\varepsilon_{ij} = \frac{1}{2\mu} \sigma_{ij} - \frac{\lambda}{2\mu(3\lambda+2\mu)} \sigma_{kk} \delta_{ij}$$

$$\text{Donos } C = \frac{\lambda}{2\mu(3\lambda+2\mu)}$$

$$\left[\varepsilon_{ij} = \frac{1}{2\mu} \sigma_{ij} - C(\sigma_{xx} + \sigma_{yy}) \delta_{ij} \right]$$

$$\varepsilon_{xy} = \frac{1}{2\mu} \sigma_{xy} - C(\sigma_{xx} + \sigma_{yy}) \delta_{xy} \xrightarrow{\delta_{xy}=0} \frac{1}{2} \sigma_{xy} \rightarrow \frac{1}{2} \mu \cdot \frac{3}{4} g (c^2 - y^2) x$$

$$\varepsilon_{xz} = \frac{1}{2\mu} \sigma_{xz} \rightarrow \frac{1}{2} \mu \cdot 0 \rightarrow 0$$

$$\varepsilon_{yz} = \frac{1}{2\mu} \sigma_{yz} \rightarrow \frac{1}{2} \mu \cdot 0 \rightarrow 0$$

$$\varepsilon_{xx} = \frac{1}{2\mu} \sigma_{xx} - C(\sigma_{xx} + \sigma_{yy})$$

$$\varepsilon_{yy} = \frac{1}{2\mu} \sigma_{yy} - C(\sigma_{xx} + \sigma_{yy})$$

$$\varepsilon_{zz} = \frac{1}{2\mu} \sigma_{zz} - C(\sigma_{xx} + \sigma_{yy}) \xrightarrow{\sigma_{zz}=0} -C(\sigma_{xx} + \sigma_{yy})$$

c) Verificar compatibilidad de deformaciones $\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} = 2 \frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y}$

Primero derivamos σ_{xx} y σ_{yy} porque ε_{xx} , ε_{yy} y ε_{xy} dependen de σ_{xx} y σ_{yy}

$$k = \frac{3}{4} \frac{g}{c^3}$$

$$\sigma_{xx} = k \left[\left(L^2 - \frac{2}{3} c^2 - x^2 \right) y + \frac{2}{3} y^3 \right] \quad \sigma_{yy} = -k \left(\frac{1}{3} y^3 - c^2 y + \frac{2}{3} c^3 \right)$$

$$\frac{\partial \sigma_{xx}}{\partial y} = k \left[\left(L^2 - \frac{2}{3} c^2 - x^2 \right) + 2y^2 \right] \quad \frac{\partial \sigma_{yy}}{\partial x} = 0$$

$$\left[\frac{\partial^2 \sigma_{xx}}{\partial y^2} = k 4y \right]$$

$$\left[\frac{\partial^2 \sigma_{yy}}{\partial x^2} = 0 \right]$$

$$\frac{\partial \sigma_{xx}}{\partial x} = k \cdot -2xy$$

$$\frac{\partial \sigma_{yy}}{\partial y} = -k(y^2 - c^2)$$

$$\left[\frac{\partial^2 \sigma_{xx}}{\partial x^2} = -2ky \right]$$

$$\left[\frac{\partial^2 \sigma_{yy}}{\partial y^2} = -k 2y \right]$$

$$\sigma_{xy} = -k(c^2 - y^2)x$$

$$\frac{\partial \sigma_{xy}}{\partial x} = -k(c^2 - y^2)$$

$$\left[\frac{\partial^2 \sigma_{xy}}{\partial x \partial y} = -k(-2y) \right]$$

Reemplazamos

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} = 2 \frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y}$$

ladder:

$$\textcircled{1} \quad \frac{\partial^2}{\partial y^2} \left(\frac{1}{2u} \sigma_{xx} - C(\sigma_{xx} + \sigma_{yy}) \right) \rightarrow \frac{1}{2u} \left[\frac{\partial^2 \sigma_{xx}}{\partial y^2} - C \left(\frac{\partial^2 \sigma_{xx}}{\partial y^2} + \frac{\partial^2 \sigma_{yy}}{\partial y^2} \right) \right]$$

$$\frac{1}{2} u k 4y - C \cdot [k 4y + -k 2y] \rightarrow \frac{1}{2} u [k 4y] - C [2ky] \rightarrow \frac{4ky}{2} - C 2ky$$

$$\textcircled{2} \quad \frac{\partial^2}{\partial x^2} \left(\frac{1}{2u} \sigma_{yy} - C(\sigma_{xx} + \sigma_{yy}) \right) \rightarrow \frac{1}{2u} \left[\frac{\partial^2 \sigma_{yy}}{\partial x^2} - C \left(\frac{\partial^2 \sigma_{xx}}{\partial x^2} + \frac{\partial^2 \sigma_{yy}}{\partial x^2} \right) \right]$$

$$\frac{1}{2} u \cdot [0] - C \cdot [-2ky + 0] \rightarrow C 2ky$$

$$\textcircled{1} + \textcircled{2} = \frac{4ky}{2} - C 2ky + C 2ky = \frac{2ky}{u}$$

ladder:

$$2 \cdot \frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y} = 2 \cdot \frac{1}{2u} \left(\frac{\partial^2 \sigma_{xy}}{\partial x \partial y} \right) \rightarrow \frac{1}{u} \cdot 2ky \rightarrow \frac{2ky}{u}$$

$$\textcircled{1} + \textcircled{2} = \textcircled{3}$$

$$\frac{2ky}{u} = \frac{2ky}{u} \quad \checkmark$$

$$3) \quad V_x = k \frac{\partial \theta}{\partial y} \quad V_y = -k \frac{\partial \theta}{\partial x} \quad V_z = 0 \quad \theta(x, y)$$

irrotacional $\nabla \times V = 0$

definición de irrotacional $\nabla \times V =$

$$\begin{bmatrix} e_x & e_y & e_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{bmatrix} \begin{bmatrix} e_x & e_y \\ V_x & V_y \end{bmatrix}$$

$$\left[\left(e_x \frac{\partial}{\partial y} V_z \right) + \left(e_y \frac{\partial}{\partial z} V_x \right) + \left(e_z \frac{\partial}{\partial x} V_y \right) \right] - \left[\left(V_x \frac{\partial}{\partial y} e_z \right) + \left(V_y \frac{\partial}{\partial z} e_x \right) + \left(V_z \frac{\partial}{\partial x} e_y \right) \right]$$

$$\left(e_y \frac{\partial}{\partial z} V_x \right) + \left(e_z \frac{\partial}{\partial x} V_y \right) - \left(V_x \frac{\partial}{\partial y} e_z \right) - \left(V_y \frac{\partial}{\partial z} e_x \right)$$

$$e_x \left(-\frac{\partial}{\partial z} V_y \right) + e_y \left(\frac{\partial}{\partial z} V_x \right) + e_z \left(\frac{\partial}{\partial x} V_y - \frac{\partial}{\partial y} V_x \right)$$

$$e_x \left(-\frac{\partial}{\partial z} \left(-k \frac{\partial \theta}{\partial x} \right) \right) + e_y \left(\frac{\partial}{\partial z} \left(k \frac{\partial \theta}{\partial y} \right) \right) + e_z \left(\frac{\partial}{\partial x} \left(-k \frac{\partial \theta}{\partial y} \right) - \frac{\partial}{\partial y} \left(k \frac{\partial \theta}{\partial x} \right) \right)$$

$\underbrace{\quad}_{\substack{\text{porque } \theta \text{ no} \\ \text{depende de } z}} \quad \quad \quad \underbrace{\quad}_{\substack{\text{porque } \theta \text{ no} \\ \text{depende de } z}}$

$$e_z \left(\frac{\partial}{\partial x} \frac{\partial \theta}{\partial y} \cdot k - \frac{\partial}{\partial y} \frac{\partial \theta}{\partial x} \cdot k \right)$$

$$e_z \left(-k \frac{\partial^2 \theta}{\partial x^2} - k \frac{\partial^2 \theta}{\partial y^2} \right) \quad \text{No es igual a 0 ya que } \neq 0$$

Incompresible $\text{div}(V) = 0 \rightarrow \nabla \cdot V$

derivada

$$\nabla \cdot V = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \rightarrow \frac{\partial}{\partial x} k \frac{\partial \theta}{\partial y} - \frac{\partial}{\partial y} k \frac{\partial \theta}{\partial x} + 0$$

$$k \frac{\partial}{\partial x} \frac{\partial \theta}{\partial y} - k \frac{\partial}{\partial y} \frac{\partial \theta}{\partial x} = 0 \quad \text{si es incompresible}$$

Tarada deformación: $V_{ij} = \frac{1}{2} \left(\frac{\partial V_i}{\partial x_j} + \frac{\partial V_j}{\partial x_i} \right)$

$$V_{xx} = \frac{1}{2} \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_x}{\partial x} \right) = \frac{1}{2} \left(2 \frac{\partial V_x}{\partial x} \right) = \frac{\partial V_x}{\partial x} \rightarrow \frac{\partial V_x}{\partial x} = k \frac{\partial^2 \theta}{\partial x \partial y}$$

$$V_{yy} = \frac{\partial V_y}{\partial y} = -k \frac{\partial^2 \theta}{\partial x \partial y}$$

$$V_{zz} = \frac{\partial V_z}{\partial z} = 0$$

$$V_{xy} = \frac{1}{2} \left(\frac{\partial V_x}{\partial y} + \frac{\partial V_y}{\partial x} \right) = \frac{1}{2} \left(k \frac{\partial^2 \theta}{\partial y^2} - k \frac{\partial^2 \theta}{\partial x^2} \right) = \frac{k}{2} \left(\frac{\partial^2 \theta}{\partial y^2} - \frac{\partial^2 \theta}{\partial x^2} \right)$$

$$V_{xz} = \frac{1}{2} \left(\frac{\partial V_x}{\partial z} + \frac{\partial V_z}{\partial x} \right) = 0$$

$$V_{yz} = \frac{1}{2} \left(\frac{\partial V_y}{\partial z} + \frac{\partial V_z}{\partial y} \right) = 0$$

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Ej 4) Partiendo de la ec. del balance de cantidad de movimiento angular demostrar que el tensor de Tensiones debe ser simétrico considerando la c.d. de momento y conservación de masa

$$\text{Cont de movimiento angular: } H_i = \int_V \epsilon_{ijk} X_j P_k dV \quad (1)$$

$$\text{Momento de fuerzas externas } L_i = \int_V \epsilon_{ijk} X_j b_k dV + \int_S \epsilon_{ijk} X_j T_k dS \quad (2)$$

$$\text{Balance: } \frac{DH_i}{Dt} = L_i \quad (3)$$

(1) y (2) en (3)

$$\frac{D}{Dt} \left(\int_V \epsilon_{ijk} X_j P_k dV \right) = \int_V \epsilon_{ijk} X_j b_k dV + \int_S \epsilon_{ijk} X_j T_k dS \quad \begin{array}{l} \text{materia derivada por} \\ \text{conservación de masa} \end{array}$$

$$* \int_V \rho \frac{D}{Dt} (\epsilon_{ijk} X_j P_k) dV = \int_V \epsilon_{ijk} X_j b_k dV + \int_S \epsilon_{ijk} X_j T_k dS \quad \text{aplicamos regla del producto}$$

$$\epsilon_{ijk} \left[\left(\frac{D X_j}{Dt} \right) P_k + X_j \left(\frac{D P_k}{Dt} \right) \right] \quad \left[\frac{D X_j}{Dt} = V_j \text{ derivada de la pos respecto al tiempo} \right]$$

$$\epsilon_{ijk} V_j P_k + \epsilon_{ijk} X_j \left(\frac{D P_k}{Dt} \right) \rightarrow$$

$$V \times V + \epsilon_{ijk} X_j \left(\frac{D P_k}{Dt} \right) \rightarrow \quad V \times V = 0$$

Volviendo a *

$$\int_V \rho \left(\frac{D P_k}{Dt} \right) \epsilon_{ijk} X_j dV = \int_V \epsilon_{ijk} X_j b_k dV + \int_S \epsilon_{ijk} X_j T_k dS$$

Usamos Cauchy $T_i = \sigma_{ij} m_j \rightarrow T_k = m_l \sigma_{lk}$ en último término

$$\int_V \rho \left(\frac{D P_k}{Dt} \right) \epsilon_{ijk} X_j dV = \int_V \epsilon_{ijk} X_j b_k dV + \int_S \epsilon_{ijk} X_j \sigma_{lk} m_l dS$$

Aplicamos Gauss al último Término $\int_S A_i n_i ds = \int_V \frac{\partial A_i}{\partial x_i} dv$

$$\int_V P\left(\frac{Dv_k}{Dt}\right) \epsilon_{ijk} X_j dv = \int_V \epsilon_{ijk} X_j b_k dv + \int_V \frac{\partial}{\partial x_i} (\epsilon_{ijk} X_j \theta_{ik}) dv \quad \text{aplicamos regla del producto}$$

$$\int_V P\left(\frac{Dv_k}{Dt}\right) \epsilon_{ijk} X_j dv = \int_V \epsilon_{ijk} X_j b_k dv + \int_V \epsilon_{ijk} \left[\frac{\partial X_j}{\partial x_i} \theta_{ik} + X_j \frac{\partial \theta_{ik}}{\partial x_i} \right] dv \quad \text{reparametrizo términos}$$

$\epsilon_{ijk} \frac{\partial X_j}{\partial x_i} \theta_{ik} \rightarrow \epsilon_{ijk} \theta_{ik}$

$$\int_V P\left(\frac{Dv_k}{Dt}\right) \epsilon_{ijk} X_j dv = \int_V \epsilon_{ijk} X_j b_k dv + \int_V \epsilon_{ijk} \theta_{jk} dv + \int_V \epsilon_{ijk} X_j \frac{\partial \theta_{ik}}{\partial x_i} dv$$

$$\int_V \epsilon_{ijk} X_j P\left(\frac{Dv_k}{Dt}\right) dv - \int_V \epsilon_{ijk} X_j b_k dv - \int_V \epsilon_{ijk} \theta_{jk} dv - \int_V \epsilon_{ijk} X_j \frac{\partial \theta_{ik}}{\partial x_i} dv = 0$$

$$\int_V \epsilon_{ijk} X_j \left[P \frac{Dv_k}{Dt} - b_k - \frac{\partial \theta_{ik}}{\partial x_i} \right] dv - \int_V \epsilon_{ijk} \theta_{jk} dv = 0$$

$\hookrightarrow$ ϵ demostrando $P \frac{Dv_k}{Dt} = \frac{\partial \theta_{ik}}{\partial x_i} + b_i \rightarrow \left[P \frac{Dv_k}{Dt} - b_i - \frac{\partial \theta_{ik}}{\partial x_i} = 0 \right]$

$$\int_V \epsilon_{ijk} X_j [0] dv - \int_V \epsilon_{ijk} \theta_{jk} dv = 0$$

$$- \int_V \epsilon_{ijk} \theta_{jk} dv = 0 \quad \text{Porque esta integral de 0, lo que es lo mismo decir que } \epsilon_{ijk} \theta_{jk} = 0$$

$\epsilon_{ijk} \theta_{jk} = 0$ Tomamos $i=1$ $j,k=2,3$ no se repiten

$\epsilon_{123} \theta_{23} + \epsilon_{132} \theta_{32}$ $\epsilon_{132} = -\epsilon_{123}$ por permutación de índices

$$\epsilon_{123} \theta_{23} - \epsilon_{132} \theta_{32} = 0$$

$$\epsilon_{123} \theta_{23} = \epsilon_{123} \theta_{32}$$

$$\theta_{23} = \theta_{32} \quad \checkmark$$

Demostremos que debe ser simétrico

Recuperatorio Parcial 2 2024

5) $X_1 = a_1 e^{-t}$ $X_2 = a_2 e^t$ $X_3 = a_3 + a_2(e^{-t} - 1)$

El campo vectorial A_i está dado por

$$A_1(X_1, X_2, X_3, t) = e^{-t} (X_1 - 2X_2 + 3X_3)$$

$$A_2(X_1, X_2, X_3, t) = e^{-2t} (X_1 - 2X_2 + 3X_3)$$

$$A_3(X_1, X_2, X_3, t) = e^{-3t} (X_1 - 2X_2 + 3X_3)$$

Calcule la derivada material A_i en ese medio en el instante final a cualquier subzona

Formula Derivada material: $\frac{DA_i}{Dt} = \frac{\partial A_i}{\partial t} + \frac{\partial A_i}{\partial X_j} V_j$

$\swarrow$ campo local $\searrow$ campo convectivo

Paso 1: Encontrar el Campo de Velocidades V_i

$$V = \left(\frac{\partial X_1}{\partial t} ; \frac{\partial X_2}{\partial t} ; \frac{\partial X_3}{\partial t} \right)$$

Dirección 1:

$$X_1 = a_1 e^{-t} \rightarrow a_1 = X_1 \cdot e^t$$

derivamos $V_1: \frac{\partial X_1}{\partial t} = -a_1 e^{-t}$

Euleriano: V en función de las X

Reemplazamos: $V_1 = -X_1 e^t \cdot e^{-t} \rightarrow -X_1$

Dirección 2:

$$X_2 = a_2 e^t \quad a_2 = X_2 e^{-t}$$

$$V_2 = \frac{\partial X_2}{\partial t} = a_2 e^t$$

reemplazamos $V_2: X_2 e^{-t} \cdot e^t \rightarrow X_2$

Dirección 3

$$X_3 = a_3 + a_2(e^{-t} - 1) \rightarrow$$

$$V_3 = \frac{\partial X_3}{\partial t} = -a_2 e^{-t}$$

reemplazamos $V_3 = -X_2 e^{-t} \cdot e^{-t} = -X_2 e^{-2t}$

$$V = (-X_1 ; X_2 ; -X_2 e^{-2t})$$

Paso 2: Calcular derivadas

$$\frac{DA}{Dt} = \frac{\partial A}{\partial t} + \left[\frac{V_1 \partial A}{\partial X_1} + \frac{V_2 \partial A}{\partial X_2} + \frac{V_3 \partial A}{\partial X_3} \right]$$

$$\frac{\partial A_1}{\partial t} = -e^{-t} (X_1 - 2X_2 + 3X_3)$$

$$\frac{V_2 \partial A_1}{\partial X_2} = X_2 \cdot e^{-t} \cdot -2$$

$$\frac{V_1 \partial A_1}{\partial X_1} = -X_1 \cdot e^{-t} \cdot 1$$

$$\frac{V_3 \partial A_1}{\partial X_3} = -X_2 \cdot e^{-t} \cdot 3 \cdot e^{-t}$$

$$\text{Sumamos } \frac{\partial A_1}{\partial t} + \frac{V_1 \partial A_1}{\partial X_1}$$

$$\frac{DA_1}{Dt} = -e^{-t} (X_1 - 2X_2 + 3X_3) + (-X_1 e^{-t} - 2X_2 e^{-t} - 3X_2 e^{-3t})$$

$$\left[\frac{DA_1}{Dt} = -2X_1 e^{-t} - 3X_3 e^{-t} - 3X_2 e^{-3t} \right]$$

$$\frac{\partial A_2}{\partial t} = -2e^{-2t} (X_1 - 2X_2 + 3X_3)$$

$$\frac{V_2 \partial A_2}{\partial X_2} = X_2 \cdot -e^{-2t} \cdot (-2)$$

$$\frac{V_1 \partial A_2}{\partial X_1} = -X_1 \cdot e^{-2t} \cdot (1)$$

$$\frac{V_3 \partial A_2}{\partial X_3} = -X_2 \cdot e^{-2t} \cdot e^{-2t} \cdot 3$$

$$\frac{DA_2}{Dt} = -2e^{-2t} (X_1 - 2X_2 + 3X_3) + (-X_1 e^{-2t} - 2X_2 e^{-2t} - 3X_2 e^{-4t})$$

$$\left[\frac{DA_2}{Dt} = -3X_1 e^{-2t} + 2X_2 e^{-2t} - 6X_3 e^{-2t} - 3X_2 e^{-4t} \right]$$

$$\frac{\partial A_3}{\partial t} = -3e^{-3t} (X_1 - 2X_2 + 3X_3)$$

$$\frac{V_2 \partial A_3}{\partial X_2} = X_2 \cdot e^{-3t} \cdot -2$$

$$\frac{V_1 \partial A_3}{\partial X_1} = -X_1 \cdot e^{-3t}$$

$$\frac{V_3 \partial A_3}{\partial X_3} = -X_2 \cdot e^{-3t} \cdot 3$$

$$\frac{DA_3}{Dt} = -3e^{-3t} (X_1 - 2X_2 + 3X_3) + (-X_1 e^{-3t} - 2X_2 e^{-3t} - 3X_2 e^{-5t})$$

$$\left[\frac{DA_3}{Dt} = -4X_1 e^{-3t} + 4X_2 e^{-3t} - 9X_3 e^{-3t} - 3X_2 e^{-5t} \right]$$

Aceleración en descomposición Euleriana

$$a_i = \frac{\partial V_i}{\partial t} + V_j \frac{\partial V_i}{\partial X_j}$$